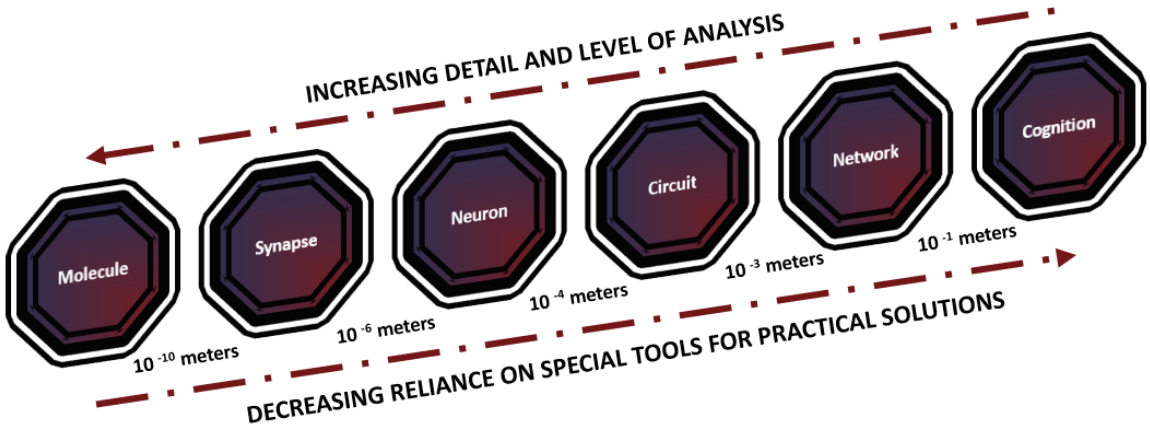
Section 7: Issues with Other Pre-Existing Models

Before juxtaposing pre-existing models, the user must remember that EvalzNow absorbs assessments at the cognitive level, which is synonymous with the functional level. As shown in the figure below, the functional level does not require special tools such as medical imaging, surgical instruments, and invasive devices to provide practical solutions to the assessment process. The need for practical solutions in the physical classroom or the technical workplace outweighs the low detail in analysis, which is evident in EvalzNow.



Aside from the signature profiles programmed into EvalzNow, users do not have consistent/derived mathematical models at the functional level that differentiate between proficiency at fixed points in time and growth – including growth projections and growth targets – across a duration of time. The absence of such a model in a traditional classroom has caused many educators to conflate proficiency and growth from year to year without any fundamental control variables. For instance, some instructors have accepted that the proportion of candidates, whom demonstrate mastery of English on a proficiency test from year to year, is interchangeable with the proportion of candidates whom have favorable growth during benchmark counting. This is a conceptual error, especially when considering the fact that time is a control variable in proficiency, but time is not a control variable in growth. In other words, growth cannot be measured if there is no time-dependent data available. The lack of time-dependent data, in turn, renders projection models useless since extrapolation and best-fit curves require data points across the duration of instruction.

Moreover, educators use scores from previous courses to create growth models, and they assume that approved curricula scale linearly, which is rarely true except for in some primary school subjects.

Consider the following:

- Candidate learned all essential grammar concepts at a 3rd-grade level.
- Candidate learned all essential grammar concepts at a 4th-grade level. **(Linear Scale with Item 1)**
- Candidate learned A1-A3 (beginner) material.
- Candidate learned B1-B3 (intermediate) material. **(Scale Not Linear with Item 3)**

Whenever there is no linear scale, the growth targets must be reset to make way for reliable data streams at the candidate level, the classroom level, and the institutional level. Of course, there are analogues to company workers; however, the rudimentary assumptions in education are not found too often on the corporate side.

To be thorough, there are a few models, most of which are not supported by a mathematical theory or an empirical study, in place that are used in traditional classrooms and learning/training platforms.

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- The student growth percentile (SGP) model determines a candidate's growth by percentile after an arbitrary relationship is determined between a collection of previous candidate scores and current candidate scores using quantile regression. Although quantile regression does not make any assumptions about the residual distribution of candidate scores, the practicality of the model is flawed because quantile regression requires a relationship between a dependent variable and independent variables to exist among candidates. Also, regression techniques do not force curve-fit equations to incorporate all data points (hence, best-fit lines in mathematics)

    - The model incorrectly assumes that one (1) candidate's growth is dependent upon the growth of other candidates in a similar classroom.
    - The model incorrectly assumes that the learning capacity of all candidates is the same. Even worse, it assumes that there is no difference in learning among a target population group, for instance, among right-brained learners, left-brained learners, cognitive learners, kinesthetic learners, and so forth. The learning capacity baseline is somehow driven by the score of the median candidate whom cannot represent all candidates in a classroom/district.
    - The model incorrectly assumes that the instructor is flawlessly teaching the course content in the same manner every time the course is offered.
    - The model does not account for any instructor-related variables, so it cannot attest to whether instructors are meeting the needs of candidates or not.
    - The model does not reflect individual growth differentials. It does not calculate how much a candidate has grown in the learning process or how much growth is necessary to achieve a target level.
    - The model has the potential for random sorting concerns and errors in percentile classification because the relationship is arbitrary. There is no way to prove that the relationship is feasible.
    - The model uses diagnostic tests, which are proficiency tests, as baselines for candidate growth, even though proficiency and growth are two (2) completely different concepts.

- The value-added model (VAM) attempts to isolate the impact that a specific instructor has on the learning capacity of all candidates. If the candidates perform well, for instance, then the outcome is attributed to the instructor and vice versa. This model is impractical because there is no way to prove that the instructor is the only variable or is the defining variable in the learning process. There are various vehicles associated with candidate learning (E.G. tutors, textbooks, educational programs, home environments, gaming, and so forth)

    - The Needs Assessment file is a source when deciding the impact that instructors have on candidate learning. With that being said, it becomes pointless to have a model like this.

- The value-table model (VTM) is an oversimplified point system that rewards institutions that either help candidates move from lower levels of proficiency to higher levels of proficiency or maintain passing proficiency scores among their candidate body.

    - The model does not consider candidate growth.
    - The model does not possess accountability standards for instructors.
    - These institutions normally administer difficult placement tests to ensure that candidates will exceed proficiency requirements prior to enrollment.

- The projection model and the trajectory model use trends to predict or to project whether a candidate will pass future proficiency tests or not. The reference point for the prediction or the projection is a past performance point.

    - The model ignores the fact that proficiency occurs at fixed points in time.
    - The model ignores the fact that past performance points might not be credible to or might not be available for a future course.
    - The model does not have any accountability standards for the instructor.

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- Diagnostic assessment for growth is used in learning platforms by coupling engagement with the outcome of a diagnostic assessment to set initial growth. Afterward, an algorithm sets a growth target by adding the initial growth and the average of accumulated outcomes for a given percentile.

    - Engagement is a tertiary factor in the learning process in comparison to proficiency and growth. Not to mention, it is inversely proportional to challenge levels. If the engagement level is high, then the challenge level is low. If the engagement level is low, then the challenge level is high. Machine learning and AI platforms are not programmed to find the minimum engagement level that leads to the maximum challenge level. There are also practical challenges. For instance, candidates might fall asleep, might spend more time on easy questions, might take breaks, and so forth.
    - The model ignores the fact that growth targets cannot be created with proficiency-based tests. A diagnostic assessment is a proficiency-based test.
    - The growth of one (1) candidate should not be based on the outcomes of other candidates, especially since candidate learning, candidate strengths, and candidate weaknesses differ.
    - A diagnostic test does not test a set of benchmarks in full. One question on a diagnostic test, for instance, can account for 2 or more benchmarks; therefore, candidate growth outcomes might either be impractical or be attributed to the wrong learning mechanisms.
    - Learning platforms duplicate questions from the diagnostic assessment in the formative assessment. This is a problem because learning platforms normally show the solutions to questions in a black-box manner, which undermines problem-solving and critical thinking. First, the candidate is given an opportunity to answer a question in the diagnostic phase, which is not graded and which provides the solutions. Then, the candidate encounters the question again, albeit with different numbers or selection shifts, in a graded state where the candidate will pass if (s)he remembers the solution process.

- The Theory of Connectivity, which was proposed by a neuroscientist named Dr. Joe Tsien, was developed on the premise that the human brain operates on a basic algorithm with broad intelligence implications. Within the theory, he proposes that complex brain computations can be reduced to the empirical equation – $N = 2^i – 1$, where N is the number of neural cliques connected in different possible ways and i is the information being received from such cliques.

    - This theory is not practical because it will require candidates to be connected to machines that monitor neural receptors during the learning process.
    - The concept has not been proven to-date, especially in dopamine-dependent neurons.
    - The model has a high degree of variability (randomness) that would require independent modeling or correction factors to the empirical equation. Dr. Tsien conceded that intelligence requires dealing with uncertainty and infinite possibilities, so it is unclear how the empirical equation can assemble neurons – systematically, adaptively, or spontaneously – to acquire knowledge.
    - The model does not account for the different mental states of humans (E.G. hungover, restless, depressed, etc.). It also does not account for the effects that energy sources, such as caffeine or B vitamins, play during the learning process.
    - The theory uses ‘2’ as the base because the cliques receive inputs (state 1) or they do not receive inputs (state 2); however, the theory does not address partial inputs or time-varying inputs.

The Theory of Connectivity requires more investigation; however, the differences between Dr. Tsien’s approach and the EvalzNow approach must be clarified. Dr. Tsien is researching knowledge acquisition across the cerebral hemispheres from a stochastic neural perspective driven by animal studies; however, the EvalzNow approach is calculating knowledge acquisition from a functional perspective driven by mutual causality of a closed system for humans. This is analogous to a heart surgeon evaluating how interconnected neurons cause cardiovascular concerns instead of how the functional units (E.G. pumps, valves, diaphragms, etc.) cause cardiovascular concerns. The neural perspective is far from pragmatic.

- Employment assessment software does not account for proficiency metrics and growth metrics during the assessment period, even though the demonstration of knowledge and the accumulation of knowledge for any job depend on such metrics. Bypassing the critical metrics of knowledge assessments leads to a subjective appraisal system that cannot determine asset value.

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- There can be a halo effect where a rater of a system weighs different aspects of an employee's performance profile incorrectly.
- The rater might feel uncomfortable rating someone low in a given category. This is evident in cases where managers empathize with the employee's personal problems, have strong working relationships, or rate based on perceived potential ability.
- Training is normally a checkmark item rather than a scored item.
- The assessment software accepts managerial inputs that might not be relevant to the key skills or the responsibilities of the job.
- The assessment software is not driven by small rubric scales and SMART goals, so the data is not reliable. For instance, two (2) different raters in the same department of the employee may have large deviations.
- The criteria within the assessment software normally generalize job activities across all departments, so the rating scale is not always authentic/applicable to certain job duties.
- Assessments for companies are normally conducted at predefined times rather than continuous intervals.
- A small fraction of the assessment software focuses on assessment. For instance, time tracking, inventory, HR resource centers, and accounting integrations have nothing to do with areas of improvement and growth targets. Not to mention, most companies have processes and procedures in place to handle these items.
- Most assessment software do not provide direct job tasks at the pre-employment stage.

- There are various neurological research/experimental techniques, surgical and non-surgical, that can be used to monitor/alter/reshape brain activity. These techniques include whole-brain imaging, animal-to-human paradigms, stereotaxic surgeries, electrophysiology, microscopy, neural structure visualization + circuitry + manipulation, DNA technology & cloning, STEM-cell culturing, and implants.

  - Any technique, which requires surgery in the learning environment or the working environment, is either impractical or restricted. For instance, implanting small chips necessitates discussions on consent, indoctrination, health, legal liabilities/risks, etc. Major surgeries, in almost all cases, will be impossible, unless the candidate experienced brain trauma or brain damage. [These are not the default/majority cases]
  - Non-invasive techniques are not practical because they will require candidates to be connected to machines/devices, and they will require constant medical oversight/support of licensed professionals. These techniques will not be feasible in the typical classroom/workplace, and they will most likely lead to high budgets.
  - Neurological research and experimental techniques neither isolate nor filter brain activity based on the defining factors of competency. Instead, the entire brain and numerous contributing factors affecting the brain are superimposed in the output results for further investigations. The time spent in these investigations, going candidate by candidate, will unacceptably dampen the effort, especially for large campuses or for large corporate enterprises.
  - These techniques will face obstacles in the pass/fail criteria of the educational system, which is polarized on proficiency and growth. Neuroscience provides circumstantial evidence rather than direct evidence, so it will be difficult to attribute density changes, volume changes, no response, dormancy, hyperactivity, etc. of brain areas definitively to either proficiency or growth from a policy perspective.

Although the aforementioned neurological research/experimental techniques can provide a deeper understanding of brain behavior/activity (E.G. molecular) than EvalzNow (E.G. functional), these neurological research/experimental techniques cannot be implemented pragmatically/practically in a typical classroom or workplace as EvalzNow can be. Put in another way, EvalzNow is the optimal approach to the learning process and the human capital assessment process when considering time, cost, practicality, health & safety, level of expertise, implementation, and any other risk. (Please refer to the diagram at the beginning of this section.)

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